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Windows 11 Security Walkthrough — Self-Contained Guide.....4

A step-by-step guide for checking the security of any Windows 11 home PC, including how to spot and remove the AppSuite/TamperedChef malware family, harden privacy and account security, and prevent browsers from auto-restarting at boot. Based on procedures used on real machines. Applicable to a spouse's, parent's, or any home Windows PC.

Written so you can follow it on a separate computer without help.

2026 Time-Sensitive Update — Read This First

Two things make mid-2026 a particularly important time to run through this guide:

- **Secure Boot certificate expiration:** The certificates that underpin Secure Boot on most Windows PCs since 2011 begin expiring in late June 2026. Microsoft has been pushing updated certificates via Windows Update. Make sure Windows Update is fully current (Step 1 below) so your PC receives these automatically — this is largely invisible if updates are current, but PCs that have been neglected for updates could see Secure Boot issues down the road.
- **A major June 2026 patch:** Microsoft shipped one of its largest security updates ever in early June 2026. If your PC shows “Restart required” in Windows Update, don't put it off — restart and let it complete.

Also new: if you set up a new Windows 11 PC recently and it has a TPM 2.0 chip, Windows may have already turned on device encryption automatically — even with a local (non-Microsoft) account. If Step 3 below

shows encryption already On, that's expected and good; just make sure the recovery key is backed up as described.

Before you start

Have ready:

- Phone with internet (used as the “trusted second device”)
- Pen and paper (or a notes app on your phone) to write down what you find
- Approximately 1–3 hours uninterrupted (longer if malware is found)
- The PC's user knowing their Microsoft account password and PIN

Important rule throughout:

If you find anything suspicious or unfamiliar, don't act immediately — write down what you saw and consult either this guide or someone knowledgeable. Some things look bad but are normal; some things look normal but are bad. The wrong “fix” can break a working system.

Quick-reference settings table (target states)

For every setting touched in this guide, here's what it should be set to. Use this as a checklist.

Setting	Location	Desired state
Windows version	Settings → System → About	Windows 11, version 23H2 / 24H2 / 25H2
Windows Update status	Settings → Windows Update	“You're up to date”
Receive updates for other Microsoft products	Windows Update → Advanced options	On
Get me up to date	Windows Update → Advanced options	On
Notify me when	Windows Update → Advanced	On

a restart is required	options	
Windows Security tiles (all 6)	Windows Security home	Green check / “No action needed”
Real-time protection	Virus & threat protection → Manage settings	On
Cloud-delivered protection	Virus & threat protection → Manage settings	On
Tamper Protection	Virus & threat protection → Manage settings	On (critical — malware tries to disable this)
Automatic sample submission	Virus & threat protection → Manage settings	On
Controlled folder access	Virus & threat protection → Manage settings	On if technically comfortable; otherwise leave default Off
Defender folder exclusion (anti CPU loop)	Virus & threat protection → Exclusions	Add C:\ProgramData\Microsoft\Windows Defender (only if Defender is pegging CPU)
Firewall — Domain network	Firewall & network protection	On
Firewall — Private network	Firewall & network protection	On
Firewall — Public network	Firewall & network protection	On
Reputation-based protection	App & browser control	On (all sub-toggles On)
Exploit protection	App & browser control	Default settings (don’t change unless you know why)
Core isolation / Memory integrity	Device security → Core isolation	On if available without breaking drivers
Device encryption / BitLocker	Settings → Privacy & security → Device encryption	On
BitLocker	account.microsoft.com/devices/	Yes, in 2+ places, NOT on

recovery key backed up	recoverykey + offline copy	the encrypted drive
Windows Hello PIN	Settings → Accounts → Sign-in options	Set up
“Only allow Windows Hello sign-in for Microsoft accounts”	Sign-in options (bottom)	On
Picture password	Sign-in options	Off / not set up
Dynamic lock	Sign-in options	User preference (Off is fine)
Other users on the PC	Settings → Accounts → Other users	Only people who actually use this PC; no unfamiliar entries
Two-step verification (Microsoft account)	account.microsoft.com/security	On
Microsoft account recovery code	Advanced security options → Recovery code	Saved offline alongside BitLocker key
App install source	Settings → Apps → Advanced app settings → Choose where to get apps	“Anywhere, but warn me” or stricter (NOT “Anywhere” silent)
Chrome “Offer to save passwords”	Chrome Settings → Autofill → Google Password Manager → Settings	Off (use a real password manager instead)
Chrome “Auto Sign-in”	Same page	Off
Browser default search engine	Browser Settings → Search engine	Google / Bing / DuckDuckGo (NOT unfamiliar names)
Browser homepage / startup pages	Browser Settings → On startup	“New Tab page” or “Continue where you left off”
Edge “Startup boost”	Edge → Settings → System and performance	Off (prevents auto-launch at boot)
Edge “Continue running background extensions and apps when	Edge → Settings → System and performance	Off

Microsoft Edge is closed		
Edge “Sleeping tabs” / Efficiency mode	Edge → Settings → System and performance	On (optional, saves RAM)
Chrome “Continue running background apps when Google Chrome is closed”	Chrome → Settings → System (chrome://settings/system)	Off
Chrome “Memory Saver”	Chrome → Settings → Performance (chrome://settings/performance)	On (Maximum on ≤ 8 GB systems)
Default browser	Settings → Apps → Default apps	Set to your preferred browser; click “Set default” at the top
Windows Widgets	Right-click taskbar → Taskbar settings → Widgets	Off (stops Edge background processes)
Microsoft Edge in Windows Startup apps	Task Manager → Startup apps tab	Disabled
Edge scheduled tasks	Task Scheduler → Task Scheduler Library	MicrosoftEdgeUpdate* tasks Disabled
Email forwarding rules	Gmail: gear → See all settings → Forwarding	None unless user set them
Email filters	Gmail: gear → See all settings → Filters	None unless user set them
2FA on every important account	Each site’s Security settings	On, authenticator app preferred
Recovery email on important accounts	Each site’s profile / security settings	An active address the user controls (NOT a closed Yahoo, etc.)

What to Keep Running vs. What to Turn Off (Microsoft Services)

Windows 11 bundles a lot of Microsoft services and background features. Some are core to security and should always stay on; others are conveniences you can safely turn off if you don't use them, especially to save RAM on lower-spec PCs. This table is a fast reference — details for each item are covered in the relevant step.

Service / Feature	Keep On / Turn Off	Why
Windows Update	Keep On	Core to security. Never disable, including the related Update Orchestrator service.
Microsoft Defender (all components)	Keep On	Real-time protection, Tamper Protection, SmartScreen, firewall — all should remain on. See Step 2.
Device encryption / BitLocker	Keep On	Protects your data if the device is lost or stolen. Back up the recovery key (Step 3) and never turn this off.
Windows Hello (PIN / biometrics)	Keep On	TPM-bound sign-in is stronger than a typed password alone. See Step 4.
Windows Widgets	Safe to Turn Off	Pure convenience feature (news/weather panel). Runs background Edge processes even when unused. See Step 6, Part F.
Edge “Startup boost” & background apps	Safe to Turn Off	Convenience-only; pre-launches Edge at boot and keeps it running after close. Turning off saves RAM. See Step 6, Part D.
Edge update scheduled tasks	Safe to Disable	Edge itself still updates normally through Windows Update / its own update channel if you use Edge. Disabling just stops it from resurrecting background relaunched. See Step 6, Part H.
Browser built-in password managers (Chrome/Edge/Firefox)	Turn Off	Convenient but a single point of failure if the PC is compromised. Use a dedicated password manager instead (see Step 6 Part I and Phase 5).
Cortana / Copilot, Game Bar, OneDrive auto-sync (if unused)	Optional — Turn Off if Unused	Not security risks by themselves, but each is a background process. Disabling unused ones frees RAM, especially on 8 GB systems (Phase 5 — Performance hygiene).
Diagnostic data / advertising ID (Settings → Privacy)	Safe to Turn Off	Privacy preference, not a security requirement either way. Settings → Privacy & security → Diagnostics & feedback → set to “Required diagnostic data” and turn off

		personalized advertising.
Remote Desktop / Remote Assistance (if not used)	Turn Off	Settings → System → Remote Desktop. If you don't use it, an open remote-access path is unnecessary attack surface.

Phase 1: Baseline Security Checks

Step 1 — Windows Update

Why: The single biggest factor in PC security. Patches close known vulnerabilities.

1. Press Windows key → type Windows Update → press Enter.
2. Read the status at the top:
 - “You’re up to date” = good
 - “Updates available” or “Restart required” = click Download/Install or Restart, then come back to this step
 - “Checking for updates...” = wait
1. Click Check for updates to be sure.
2. Click Advanced options and confirm these are On:
 - Receive updates for other Microsoft products
 - Get me up to date
 - Notify me when a restart is required
1. Check Windows version: Settings → System → About. Confirm:
 - **Edition:** Windows 11 Home or Pro
 - **Version:** 23H2, 24H2, or 25H2 (current as of 2026)
 - **NOT Windows 10:** Windows 10 reached end of free security updates in October 2025. Without Extended Security Updates (ESU), it's no longer getting patches. Strong recommendation: upgrade to Windows 11 if hardware supports it.

Optional updates note:

Optional preview cumulative updates (e.g., “2026-04 Cumulative Update Preview for Windows 11”) should be skipped. They become required updates the following month and occasionally ship with bugs that get fixed before release. Driver updates: install only if you’re having a related hardware issue.

Step 2 — Windows Security (Defender)

Why: Built-in antivirus, firewall, threat protection. Generally excellent on Windows 11; most home users don’t need a third-party AV.

1. Press Windows key → type Windows Security → press Enter.
2. Each of these six tiles must show a green check or “No action needed”:

- Virus & threat protection
- Account protection
- Firewall & network protection
- App & browser control
- Device security
- Device performance & health

If any shows yellow ! or red X: click into it, read the warning, fix the specific issue it names. Don’t move on until all green.

1. Click Virus & threat protection → Manage settings. Set these to On:

- Real-time protection → On
- Cloud-delivered protection → On
- Automatic sample submission → On
- Tamper Protection → On (critical — malware tries to disable this; if it’s Off and the user didn’t disable it, that’s a red flag)
- Controlled folder access → optional. Turn On if comfortable troubleshooting blocked apps; otherwise leave at default.

1. Click Firewall & network protection. All three must be On:

- Domain network → On
- Private network → On
- Public network → On

If any are Off, click into that network and toggle Microsoft Defender Firewall to On.

1. Click App & browser control → Reputation-based protection settings. All sub-toggles should be On:
 - Check apps and files → On
 - SmartScreen for Microsoft Edge → On
 - Phishing protection → On (all three sub-options On)
 - Potentially unwanted app blocking → On (both Block apps and Block downloads On)
 - SmartScreen for Microsoft Store apps → On
1. Click Device security → if Core isolation is shown, click into it. Memory integrity should be On if compatible. If it says incompatible drivers are blocking it, leave Off (forcing it can break boot); user can revisit after driver updates.

Note on Memory Integrity overhead: Memory Integrity uses Windows' Virtualization-Based Security (VBS) to isolate critical kernel processes in a hypervisor-protected VM. This is what spawns vmmem and vmwp processes (typically 150–300 MB RAM). It is legitimate, valuable security — leave it On.

1. Check for third-party antivirus in Settings → Apps → Installed apps. Search for: Norton, McAfee, Avast, AVG, Avira, Kaspersky, Bitdefender, ESET, Webroot. Desired state: zero third-party AVs installed (Defender is enough).
 - Pre-installed trial AV (came with the PC) → uninstall, Defender is enough
 - Multiple AVs → uninstall all but one, ideally just keep Defender
 - Avira specifically → see the dedicated Avira removal procedure below

Defender CPU-loop fix (only if needed): If Antimalware Service Executable (MsMpEng.exe) is consistently using high CPU at idle,

Defender may be scanning its own log files in a loop. Add a folder exclusion: Windows Security → Virus & threat protection → Manage settings → Add or remove exclusions → Add an exclusion → Folder → C:\ProgramData\Microsoft\Windows Defender. Don't add this exclusion preemptively — only if Defender is misbehaving.

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Step 3 — Device Encryption / BitLocker

Why: Protects data if the PC is lost or stolen. Without it, anyone with physical access can read the drive.

1. Settings → Privacy & security → Device encryption.
2. Device encryption → On. If Off, toggle it On (encryption will run in the background; PC stays usable).
3. If “Device encryption” page doesn't exist: search Start menu for Manage BitLocker. Each drive (at least the OS drive C:) → BitLocker On.
4. Find and save the recovery key (this is critical — losing it means losing all data if Windows ever can't unlock the drive):
 - On a phone or another device, go to account.microsoft.com/devices/recoverykey
 - Sign in with the Microsoft account used on this PC
 - Should be: a 48-digit recovery key (eight blocks of six digits) listed for this device
 - Save it in two places: (a) print and store with important papers, OR password manager entry; AND (b) USB drive stored physically separate from the laptop
 - Must NOT save it only on the encrypted drive itself — chicken-and-egg problem when you'd actually need it
1. If the recovery key page is empty / no key shown: search Start menu for Manage BitLocker → click Back up your recovery key → choose Save to file (USB), Print, or Save to Microsoft account. Then re-check account.microsoft.com/devices/recoverykey to confirm it uploaded.

Step 4 — User accounts & sign-in

1. Settings → Accounts → Your info:

- Should show: a Microsoft account (email displayed) — needed for BitLocker key auto-backup. Local account works but loses that benefit.
- Account type: Administrator is typical and acceptable for a single-user home PC. Best practice (optional) is to make a second Standard user for daily use and only sign in to the admin account when installing software — biggest single reduction in malware blast radius. Skip if it adds too much friction.

1. Settings → Accounts → Sign-in options:

- Windows Hello PIN → Set up (6+ digits). The PIN is bound to the device's TPM, so it can't be brute-forced remotely.
- Windows Hello Face / Fingerprint → Set up if hardware supports it (laptops with IR cameras / fingerprint readers). If "Unavailable," that's a hardware limitation — no problem.
- Password → keep set up as fallback, but PIN should be the daily sign-in.
- Picture password → Off / not set up. Less secure than PIN; skip.
- At the bottom: "For improved security, only allow Windows Hello sign-in for Microsoft accounts on this device" → On.
- Dynamic lock (locks PC when paired phone leaves Bluetooth range) → user preference. Off is fine.

1. Settings → Accounts → Other users and Family:

- Should show: only people who actually use this PC.
- If anything is unfamiliar (a name/email you don't recognize) → that's a red flag, do not delete yet, ask the user first.
- Cleanest state: zero "Other users" if it's a single-user PC.

1. Two-step verification on the Microsoft account (do from phone):

- Go to account.microsoft.com/security
- Sign in
- Two-step verification → On. If currently Off, turn it on now.

- Methods to set up: at least two. In order of preference: (a) Microsoft Authenticator app, (b) phone SMS, (c) email. Authenticator app should be primary.
 - Recovery code → Generate and Save. account.microsoft.com/security → Advanced security options → Recovery code → Generate. Save with the BitLocker key (print, password manager, USB). Without this code, losing the phone can lock you out permanently.
-

Phase 2: Apps Audit (where malware is usually found)

This is the most important part for catching malware infections.

Step 5 — Identify suspicious installed apps

1. Tighten install source first. Settings → Apps → Advanced app settings → Choose where to get apps:
 - Should be: “Anywhere, but warn me before installing an app that’s not from the Microsoft Store” or stricter (“Microsoft Store only”).
 - NOT recommended: “Anywhere” (silent — no warnings).
1. Settings → Apps → Installed apps.
2. Sort by Install date, newest first. (This catches anything that snuck in recently.)
3. Desired state: every app is one the user recognizes and intentionally installed.
4. Things to look for and remove:
 - Apps the user doesn’t recognize
 - Generic-named apps with no brand: “PDF Editor,” “Video Converter,” “Music Player,” “PC Cleaner”
 - PUPs by reputation (uninstall on sight): Fortect, Reimage, MyCleanPC, Advanced SystemCare, Driver Updater, WinZip Driver Updater, PC Optimizer Pro, Driver Booster
 - Multiple toolbars or “search helpers”
 - Old AV trials never activated (Norton, McAfee, etc.)

1. For anything unfamiliar, find the publisher name first (don't uninstall yet — a generic app from a major publisher is often legitimate bundled software):
 - In Installed apps, click the app entry → Advanced options (Store apps), or
 - Control Panel → Programs and Features (always shows publisher in a column — most reliable for traditional installers), or
 - PowerShell (right-click Start → Terminal): `Get-AppxPackage *suspicious_name*` | Select Name, Publisher

CRITICAL — Known malicious publishers and apps to remove on sight

If you see ANY of these, treat as malware. Do not run them. Skip to Phase 3: Incident Response below before uninstalling.

Publisher / Name	What it is
AppSuite (any product)	TamperedChef backdoor family
PDF Editor 1.x.xx by AppSuite	Confirmed backdoor
ManualFinder	Same malware family
OneLaunch	Same malware family
Wave Browser / Wave	Same malware family
Shift browser	Same malware family
Generic “PDF Editor” with no recognizable brand	Often this family

These are SEO'd via search ads for things like “free PDF editor.” Multiple security firms (Microsoft Defender, Malwarebytes, G DATA, TrueSec) classify them as full backdoors capable of password theft, data exfiltration, and remote control.

Common PUPs to remove (not malware, but unhelpful)

- Fortect (formerly Reimage) — scareware “PC repair”
- Avira Free if running alongside Defender — redundant
- Norton 360 trial / McAfee LiveSafe trial if not actively used — pre-installed bloatware
- “Driver Updater” / “Registry Cleaner” anything

Step 6 — Browser settings, extensions, and auto-restart prevention

For each browser used on the PC (Chrome, Edge, Firefox, others):

Part A — Extensions audit

Paste `chrome://extensions` / `edge://extensions` / `about:addons` into the address bar.

- Desired state: every extension recognized and installed deliberately by the user.
- Remove on sight: anything called “Coupon,” “Shopping helper,” “Deal finder,” “Search helper,” “Wave,” “Shift,” anything not recognized, anything from an unknown publisher.
- Pay attention to permissions: “Read and change all your data on all websites” is powerful — only acceptable for known good extensions (a password manager you chose deliberately, or a well-known ad blocker like uBlock Origin).
- Allow in incognito → user preference. Off by default; turn On only for password managers / ad blockers if user wants them in private windows.

Part B — Default search engine

(Settings → Search engine):

- Sh
- ould be: Google, Bing, or DuckDuckGo.
- Red flags to remove: “search-redirect,” “yahoo-search” (only legitimate “Yahoo” entry is fine), “myway,” “ask.com,” anything else unfamiliar.
- In Manage search engines and site search, delete unfamiliar entries from the list.

Part C — Homepage / startup pages

(Settings → On startup):

- Acceptable: “Open the New Tab page” or “Continue where you left off.”
- Red flag: “Open a specific page or set of pages” with a URL the user doesn’t recognize → remove that URL.

- If a site like Facebook keeps reopening when you launch the browser: it's either pinned (right-click the tab → Unpin), set as a startup page (remove from On startup list), or being launched by an extension (check Part A).

Part D — Stop browsers from auto-restarting after reboot or close

Browsers like Edge and Chrome are configured by default to silently keep running even after you close them, and to pre-launch at Windows boot.

This eats RAM (often 1–2 GB), drives “why is m

y PC slow?” problems, and resurrects browser instances even when you didn't open them. Turn this behavior off.

Microsoft Edge — stop background relaunch:

1. Open Edge → click the 3 dots in the top-right → Settings.
2. Click System and performance in the left sidebar (or paste `edge://settings/system` into the address bar).
3. Turn OFF: Startup boost (this is what pre-launches Edge silently when Windows boots).
4. Turn OFF: “Continue running background extensions and apps when Microsoft Edge is closed.”
5. Optional: Turn ON “Save resources with sleeping tabs” / Efficiency mode for additional RAM savings.
6. Close Edge fully when done.

Google Chrome — stop background relaunch:

1. Open Chrome → click the 3 dots in the top-right → Settings.
2. Click System in the left sidebar (or paste `chrome://settings/system`).
3. Turn OFF: “Continue running background apps when Google Chrome is closed.”
4. Optional: paste `chrome://settings/performance` and turn ON Memory Saver. Set to Maximum on systems with 8 GB or less RAM.
5. Close Chrome fully when done.

Note: Chrome doesn't have an exact equivalent of Edge's "Startup boost" — "Continue running background apps" is the only Chrome setting that keeps it alive after close. If Chrome is still launching at Windows boot after you turn this off, the cause is in Windows Startup apps (see Part E).

Mozilla Firefox — stop session restore on launch (if applicable):

1. Firefox → hamburger menu → Settings → General.
2. Under Startup: uncheck "Open previous windows and tabs."
3. Close Firefox fully.

Part E — Set the correct default browser (stops Windows from auto-launching Edge)

Even with Edge's background settings off, Windows itself will launch Edge whenever you click a link from Start menu search, the Widgets panel, Outlook, Teams, or notifications — unless a different browser is your system default.

1. Press Windows key → type Default apps → Enter.
2. Search for or click your preferred browser (Chrome, Firefox, etc.) in the app list.
3. Click "Set default" at the top — this assigns it for HTTP, HTTPS, HTML, and all related link types in one click.

Part F — Disable Windows Widgets (a major hidden Edge spawner)

The Widgets panel uses Edge processes in the background to fetch news, weather, and feed content — even if you never open Widgets. Turning Widgets off kills 2–3 ongoing Edge processes for most users.

1. Right-click an empty area of the taskbar → Taskbar settings.
2. Find Widgets and turn it Off.

Part G — Disable browsers in Windows Startup apps

Even with browser-side settings disabled, Windows can still auto-launch Edge or Chrome at boot if they're in the Startup apps list (separate from the browser's own settings).

1. Open Task Manager (Ctrl + Shift + Esc).
2. Click the Startup apps tab.
3. Look for: Microsoft Edge, MicrosoftEdgeUpdate, Google Chrome, GoogleUpdate, GoogleChromeAutoLaunch_*, or anything browser-related.
4. Right-click each → Disable.

Part H — Disable Edge's scheduled tasks (the persistent ones)

Edge schedules itself in Windows Task Scheduler to launch periodically for updates and “preloading,” which can keep resurrecting it.

1. Press Windows key → type Task Scheduler → Enter.
2. In the left panel, click Task Scheduler Library.
3. Look for entries with MicrosoftEdge in the name (typically MicrosoftEdgeUpdateBrowserReplacement, MicrosoftEdgeUpdateTaskMachineCore, MicrosoftEdgeUpdateTaskMachineUA).
4. Right-click each → Disable. (Don't Delete — disabling is reversible.)

Verify it worked:

1. Reboot the PC.
2. Before opening any browser, open Task Manager.
3. Sort the Processes tab by Name and look for “Microsoft Edge” and “Google Chrome” entries.
4. There should be none. If Edge still appears, you missed Part F (Widgets) or Part G (Startup apps) or Part H (scheduled tasks).

Part I — Chrome's built-in password manager

(Chrome → Settings → Autofill and passwords → Google Password Manager → Settings):

- Offer to save passwords → Off (use a dedicated password manager instead — see Phase 5 for what to look for)
- Auto Sign-in → Off

- Use a passkey or password from this device → On is fine if user uses passkeys

Part J — Chrome safe browsing

(Settings → Privacy and security → Security):

- Enhanced protection (recommended) or Standard protection → On
- Off / “No protection” → Never.

Part K — Chrome sync

(Settings → You and Google → Sync and Google services):

- Confirm sync is to the user’s actual Google account, not an unfamiliar one.
 - If sync was disrupted by malware cleanup, Sign out and sign back in to re-establish a clean session.
-

Phase 3: If You Find AppSuite / TamperedChef / Similar Malware

If Phase 2 turned up confirmed malware, switch to incident response. Do not skip steps.

Phase 3 Step 1 — Uninstall the malicious app

1. Settings → Apps → Installed apps → find the bad app → ... → Uninstall.
2. Optionally disconnect Wi-Fi during uninstall to prevent it phoning home one last time. Reconnect after.
3. Don’t trust that uninstall is complete — backdoors leave files, registry entries, and scheduled tasks. Continue to Step 2.

Phase 3 Step 2 — Run Microsoft Defender Offline scan

This is the most powerful built-in tool. It boots the PC into a clean environment and scans before Windows fully loads, catching things that hide from regular scans.

1. Save all open work and close all apps.
2. Windows Security → Virus & threat protection → Scan options (under Quick scan).
3. Select Microsoft Defender Antivirus (offline scan).
4. Click Scan now. Confirm the prompt.
5. PC reboots in ~1 minute, shows blue Windows Defender scanning screen for 10–20 minutes, then boots normally.
6. Open Windows Security → Protection history. Read every detection from this scan.
7. For each detection, confirm the action status is Quarantined or Removed. If anything says Allowed or Failed / Action needed, click the entry and take the recommended action.

Common detection names to expect for TamperedChef:

- Trojan:Win64/TamperedChef.*
- PUP.Optional.PDFEditor.*
- Registry keys under HKCU\...\Uninstall\<GUID>

Phase 3 Step 3 — Manually clean residue

Defender misses some files. Check these paths in File Explorer (paste each into the address bar):

- C:\Users\<username>\PDFEditor
- C:\Users\<username>\AppSuite
- %AppData%\PDFEditor
- %AppData%\AppSuite
- %LocalAppData%\PDFEditor
- %LocalAppData%\AppSuite
- %ProgramData%\PDFEditor
- %ProgramData%\AppSuite

For each that exists:

1. Open the folder. If it contains an Electron framework (files like chrome_*.pak, ffmpeg.dll, icudtl.dat, plus Resources/app/ or Resources/w-electron/ subfolders), it's the malware.

2. Close any File Explorer window inside that folder.
3. Navigate up one level → right-click the malware folder → Delete.
4. If “in use” / “access denied,” reboot first then delete.
5. Empty the Recycle Bin to make sure files are gone.

Re-check all paths above to confirm “location not found.”

Phase 3 Step 4 — Run Malwarebytes Free as a second-opinion scan

Different scanners catch different things. Malwarebytes is particularly good at PUPs and adware residue.

1. From the PC’s browser, go to malwarebytes.com. Verify the URL bar reads exactly that — not free-malwarebytes.com or anything similar.
2. Download Malwarebytes Free. During install, if you see a checkbox like “Enable free trial of Malwarebytes Premium” or a prompt about Browser Guard / real-time protection, decline it (uncheck the box, or click “Skip” / “No thanks”). This avoids a known conflict where Malwarebytes Premium takes over as your primary antivirus and disables Windows Defender — see the box below.

Why this matters: Windows only allows one real-time antivirus at a time. If Malwarebytes Premium (or its 14-day trial) activates, it can register itself as the primary antivirus and Windows will automatically turn off Defender’s real-time protection — leaving you with a confusing dual-AV state that’s a hassle to undo. Malwarebytes Free, used as an on-demand second-opinion scanner, never does this.

If the trial activated anyway: Open Malwarebytes → Settings → Account, and look for an option to deactivate the Premium trial — do this immediately rather than waiting for the trial to expire on its own. Then open Windows Security → Virus & threat protection and confirm **Microsoft Defender Antivirus** is listed as the active protection. If Defender still shows as off, open Malwarebytes → Settings → Security and turn off all Real-Time Protection toggles, then re-check. If Defender still won’t turn back on after that, download the “Malwarebytes Support Tool” / Clean

Uninstall Tool from Malwarebytes' official support site, run a clean removal, reboot, and reinstall Malwarebytes — declining the trial this time.

3. Install and run a default scan (15–45 minutes).
4. Quarantine all detections that come up. Modern Malwarebytes splits them into pages (malware → PUPs → PUMs → cookies). Click Quarantine on every page until it stops asking. Don't click Skip.
5. Reboot when prompted to finish removing locked items.
6. Optionally install Malwarebytes Browser Guard when offered — it blocks malicious sites and is genuinely useful.

Phase 3 Step 5 — Sign out of all sessions everywhere (from phone)

To invalidate any session cookies the malware may have stolen:

Google account:

1. Phone browser → myaccount.google.com
2. Security → Your devices → Manage all devices
3. Review the list — flag anything unfamiliar
4. Sign out every device (you'll re-sign-in fresh after)

Microsoft account:

1. Phone browser → account.microsoft.com
2. Sign in (2FA prompt)
3. Security → Manage how I sign in (or Advanced security options) → Sign me out everywhere → Continue
4. Also visit account.live.com/Activity — review recent sign-ins for anything unfamiliar

If the user has other accounts (Apple, Dropbox, Facebook, etc.), do the equivalent from the phone for each.

Phase 3 Step 6 — Check email for active compromise

For each email account the user has (Gmail, Outlook, Yahoo, etc.):

1. Recent activity / sign-in history — anything from a country/IP/device they don't recognize? If yes, this is active compromise — escalate beyond this guide.
2. Sent folder — any emails the user didn't send?
3. Trash — any password-reset emails or security alerts they didn't trigger?
4. Filters / Forwarding rules (Gmail: gear → See all settings → Filters and Blocked Addresses, plus Forwarding and POP/IMAP) — any rules they didn't create? Especially silent forwards to unknown addresses.

If everything checks out clean, the malware exposure is historical (passwords stolen, sitting in a database) rather than active (someone currently in the accounts).

Phase 3 Step 7 — Password rotation plan

Assume any password saved in Chrome (or any browser's built-in password manager) on this PC is compromised. Plan:

Tier 1 — rotate this week (~15–25 accounts):

- Email accounts (master keys)
- Microsoft account, Google account, Apple ID
- Banks, brokerage, credit card portals, PayPal/Venmo
- Tax (IRS.gov), healthcare portals, insurance
- Work accounts

Tier 2 — rotate this month (~30–60 accounts):

- Amazon, eBay, large retailers with stored cards
- Primary social accounts (especially used as recovery options)
- Anything else with payment info

Tier 3 — rotate over next 1–2 months, in batches: everything else.

For each, ask: "Have I logged in within the last 12 months?" If no, consider deleting the account instead. JustDeleteMe.xyz grades sites on deletion difficulty.

Workflow per account, ideally on phone:

1. Open the site/app → sign in with current saved password
2. Account/Security settings → Change password
3. Use a dedicated password manager (see Phase 5 for what to look for) to generate strong unique passwords. Stop using browser-built-in password storage going forward.
4. Save new password to the password manager
5. Enable 2FA if offered (authenticator app > SMS > email)
6. Sign out of all other sessions if the site offers it

Update recovery emails if the old recovery is a closed/dead account.

Phase 3 Step 8 — Disable Chrome's built-in password manager

After Tier 1+2 are migrated to a real password manager:

1. Chrome → Settings → Autofill and passwords → Google Password Manager → Settings
 2. Toggle Offer to save passwords to Off
 3. Toggle Auto Sign-in to Off
 4. Eventually delete the saved passwords from Chrome's vault entirely
-

Phase 4: Avira-Specific Removal (if applicable)

If the PC has Avira and you want to remove it (recommended if Defender is also active):

1. Settings → Apps → Installed apps → find every Avira entry (Security, Browser Safety, Phantom VPN, Privacy Pal, Software Updater, etc.)
2. Uninstall each one. Be patient — Avira's uninstaller can take 5–15 minutes per component. Watch Task Manager for CPU activity to confirm it's working, even if the window looks frozen.
3. Don't kill the uninstaller process unless there's truly no CPU activity for 5+ minutes.
4. If uninstall fails: the app may offer a Repair option. Run repair (yes, even though you want to remove it — broken installs are harder to clean up). Reboot. Then uninstall again.

5. If repair fails too: download Avira's official Registry Cleaner tool. Best path: support.avira.com → search "Support Tools" or "Registry Cleaner." Download only from avira.com (not third-party sites). Run as Administrator, reboot.
 6. After cleanup, verify Microsoft Defender is the active antivirus (Windows Security → Virus & threat protection should say "Microsoft Defender Antivirus").
-

Phase 5: Hardening Recommendations

Apply these regardless of whether malware was found.

Going-forward defensive habits

Never install free utilities from search results. Most malware arrives via SEO'd "free X tool" pages. Stick to:

- Known publishers (Adobe, Foxit, PDF24 for PDFs; Audacity for audio; VLC for video; etc.)
- Microsoft Store apps
- Edge has built-in PDF viewing/annotation that covers most needs

Use a real password manager. Stop using browser-built-in password storage. Look for one that offers:

- End-to-end encryption, so the provider itself can't read your stored passwords
- A working free tier or transparent pricing, with apps for every device you use (Windows, phone, browser extension)
- A track record of independent security audits and a clear breach-disclosure history
- Built-in passkey support and 2FA/authenticator code storage, since both are becoming standard

Several well-known options meet these criteria at the time of writing — do a quick search for “password manager comparison [current year]” to see current reviews, since this is an area where features and pricing change.

Run Malwarebytes Free monthly. Catches PUPs Defender ignores. Free version is enough — decline any Premium trial prompt, and afterward confirm Microsoft Defender is still your active antivirus (see Phase 3 Step 4 for details).

Run Defender Offline scan quarterly or any time something feels off (slow performance, unfamiliar pop-ups, browser redirects).

Enable 2FA everywhere it’s offered. Authenticator app > SMS > email. Save recovery codes for every account that issues one.

Review browser extensions every few months. They’re powerful and easy to forget.

Keep BitLocker/encryption recovery keys saved off the encrypted drive. Print, password manager, USB stick — somewhere safe and not the locked drive itself.

Keep browsers from auto-restarting. Re-verify every few months that Edge “Startup boost” and “run in background” are still off, and that your default browser is set to your preferred one. Windows updates and Edge updates have been known to silently flip these back on.

Watch for related malware family names in the future: AppSuite, ManualFinder, OneLaunch, Wave, Shift, TamperedChef. They cluster.

Performance hygiene (related to security)

A bloated, low-RAM system encourages users to disable security features for performance. Keeping the system trim helps maintain security.

- Audit Windows Startup apps quarterly: Task Manager → Startup apps. Disable anything you don’t actively use (Spotify, Steam, Discord, Adobe Creative Cloud auto-launchers, OEM bloatware like Lenovo Vantage).
- Pick one daily-driver browser. Running Chrome + Edge + Firefox simultaneously can easily eat 3–4 GB on its own.

- On 8 GB systems, set Chrome's Memory Saver to Maximum and avoid keeping Electron apps (Slack, Teams, Discord, Spotify) open when not in use — each typically consumes 300–700 MB.
- If a single Chrome or Edge instance shows multiple GB in Task Manager, close and relaunch — it's usually a leaked tab or extension.

Quick decision tree if something looks suspicious

Finding	Action
Unrecognized app from a major publisher	Probably bundled software. Note name; safe to leave for now.
Unrecognized app, generic name, unknown publisher	Search the publisher name + "PUP" or "malware". If hits, treat as malicious.
Unrecognized app named "PDF Editor" / "ManualFinder" / similar from "AppSuite"	Confirmed malware. Skip to Phase 3.
Browser default search engine is unfamiliar	Hijacked. Reset to Google/Bing in Settings, remove the unfamiliar search engine entry.
Unfamiliar device in Google "Manage devices"	Possible active compromise. Sign out, change password from phone, enable 2FA.
Unfamiliar email forwarding rule	Active compromise indicator. Remove rule, change email password, sign out everywhere, escalate.
Recent sign-in from a country the user has never been to	Same — active compromise.
Defender says "Action needed" or "Failed" on a quarantine	Click into the entry → take the suggested action.
Edge keeps relaunching after you close it	Confirm Edge System and performance toggles are off; disable Widgets; check Windows Startup apps; disable MicrosoftEdgeUpdate scheduled tasks.
Browser uses 4+ GB RAM unexpectedly	Likely a leaky tab or extension. Close browser, reopen with a fresh profile, audit extensions.

When to call for help

This guide covers most cleanup scenarios, but escalate if:

- Active compromise indicators (unfamiliar sign-ins, sent emails the user didn't send, forwarding rules they didn't set, money missing from accounts)
 - Ransomware — files renamed with weird extensions, ransom note on desktop
 - Defender Offline scan finds 50+ items or the PC behaves erratically after cleanup
 - The user has work/employer accounts on the PC — notify their IT/security team. They have routines for this.
 - The user is a current target of harassment, stalking, or domestic abuse — security tools alone aren't enough; involve professionals (TheHotline.org for DV; Coalition Against Stalkerware for stalkerware support)
-

Appendix: Specific commands you may need

PowerShell (right-click Start → Terminal)

List all installed Store apps matching a pattern:

```
Get-AppxPackage *<keyword>* | Select Name, Publisher, PackageFullName
```

List all programs (Store and traditional):

```
Get-WmiObject -Class Win32_Product | Select Name, Vendor, InstallDate
```

Force-stop and remove a specific Store app:

```
Get-AppxPackage *<exactname>* | Remove-AppxPackage
```

Identify Hyper-V-based VM workers (helpful when investigating vmmem / Memory Integrity):

```
Get-CimInstance Win32_Process -Filter "Name='vmwp.exe'" | Select-Object ProcessId, CommandLine
```

Direct browser settings URLs

Microsoft Edge:

- <edge://settings/system> — background apps and Startup boost
- <edge://settings/help> — version and updates
- <edge://extensions> — review installed extensions

Google Chrome:

- <chrome://settings/system> — background apps toggle
- <chrome://settings/performance> — Memory Saver
- <chrome://settings/onStartup> — startup pages
- <chrome://settings/searchEngines> — review search providers
- <chrome://settings/help> — version and updates
- <chrome://extensions> — review installed extensions

File Explorer paths to remember

- %AppData% → C:\Users\<user>\AppData\Roaming
- %LocalAppData% → C:\Users\<user>\AppData\Local
- %ProgramData% → C:\ProgramData
- %UserProfile% → C:\Users\<user>

Useful URLs

- account.microsoft.com/devices/recoverykey — BitLocker recovery key
- account.microsoft.com/security — Microsoft account 2FA, recovery codes
- account.live.com/Activity — Microsoft sign-in activity
- myaccount.google.com → Security — Google account, devices, password
- mysignins.microsoft.com — Microsoft work/school account
- haveibeenpwned.com — check email against known data breaches
- justdeleteme.xyz — guide to deleting unused accounts

Generated based on real cleanup procedures performed on Windows 11 Home machines, including AppSuite “PDF Editor” / TamperedChef

removal, Avira cleanup, and browser auto-restart prevention. Updated May 2026 to incorporate browser auto-restart prevention guidance.

Addendum: Firefox-Specific Hardening

Firefox uses a different settings layout than Chromium browsers (Chrome, Edge), so it gets its own walkthrough. Open Firefox and use these address-bar shortcuts (paste each into the URL bar and hit Enter). Skip this section if Firefox is not installed.

F1 — Extensions audit (about:addons)

1. Paste `about:addons` → Enter.
2. Click Extensions in the left sidebar.
3. Desired state: every extension recognized and installed deliberately by the user.
4. Remove on sight: anything called “Coupon,” “Shopping helper,” “Search helper,” “Wave,” “Shift,” unknown publishers, or extensions you don’t recognize.
5. Click Themes and Plugins as well — same rules apply.

F2 — Tracking protection and safe browsing

1. Paste `about:preferences#privacy` → Enter.
2. Under Enhanced Tracking Protection, select Strict (recommended) or Custom (only if you understand the trade-offs). Default Standard is acceptable but weaker.
3. Scroll to Deceptive Content and Dangerous Software Protection — confirm all four checkboxes are On (Block dangerous and deceptive content, dangerous downloads, unwanted/uncommon software).
4. Scroll to HTTPS-Only Mode — enable in all windows (recommended) or in private windows only. Forces secure connections wherever available.

F3 — Search engine

1. Paste `about:preferences#search` → Enter.

2. Default Search Engine: confirm it's Google, Bing, DuckDuckGo, or another known engine. Not "myway," "search-redirect," or other unfamiliar names.
3. In One-Click Search Engines (or Other Search Engines), remove any unfamiliar entries.
4. Address Bar: turn OFF "Provide search suggestions" for stricter privacy (optional).

F4 — Stop session restore on launch (auto-restart prevention)

1. Paste about:preferences → Enter.
2. Under Startup, uncheck "Open previous windows and tabs."
3. Optional: uncheck "Always check if Firefox is your default browser" to stop the recurring prompt — only if you're intentionally not using Firefox as default.

F5 — Disable telemetry and data collection

1. Paste about:preferences#privacy → Enter.
2. Scroll to Firefox Data Collection and Use.
3. Uncheck: "Allow Firefox to send technical and interaction data to Mozilla."
4. Uncheck: "Allow Firefox to install and run studies."
5. Uncheck: "Allow Firefox to send backlogged crash reports on your behalf."
6. Optional: uncheck the Firefox Suggest experimental options if shown.

F6 — Disable Pocket and recommended-content noise

1. Paste about:preferences#home → Enter.
2. Uncheck: "Recommended by Pocket."
3. Uncheck: "Sponsored stories," "Sponsored shortcuts," and "Recent activity" if you don't want them.
4. Paste about:preferences#search again — uncheck "Show search suggestions from sponsors" if present.

F7 — DNS over HTTPS (DoH)

1. Paste about:preferences#privacy → Enter.

2. Scroll to DNS over HTTPS at the bottom.
3. Recommended: Increased Protection (uses Cloudflare or NextDNS by default).
4. Max Protection only if you're comfortable with what it means (will fail closed if DoH is unavailable).

F8 — Profile audit (catches injected secondary profiles)

1. Paste `about:profiles` → Enter.
2. Confirm there's only the default profile (or profiles you intentionally created).
3. If you see an unfamiliar profile, do NOT click Remove yet — note its location and investigate. Some malware creates secondary Firefox profiles to siphon data quietly.

F9 — Update settings

1. Paste `about:preferences#general` → Enter.
2. Under Firefox Updates, leave "Automatically install updates" On.
3. Uncheck "Use a background service to install updates" if you're tight on RAM (8 GB systems) and want to reduce one persistent background service. Updates will still install — just when Firefox is running.

F10 — Built-in password manager

1. Paste `about:preferences#privacy` → Enter.
2. Scroll to Logins and Passwords.
3. Uncheck: "Ask to save logins and passwords for websites" (use a dedicated password manager instead — see Phase 5 for what to look for).
4. Click Saved Logins to review existing saved passwords. Migrate them to a real password manager and remove from Firefox.

F11 — Diagnostics if Firefox feels slow or memory-heavy

- `about:memory` — click Measure under Show memory reports to see what's using RAM.
- `about:performance` — shows per-tab and per-extension CPU and memory usage; identify the heavy tab/extension.

- [about:support](#) — Troubleshooting Information page; useful when asking for help on forums.

F12 — Optional: Multi-Account Containers extension

Mozilla’s Multi-Account Containers extension lets you isolate cookies and sessions per tab (e.g., one container for personal Google, another for work Google, a third for shopping). Reduces cross-site tracking and accidental cross-account leakage. Install from addons.mozilla.org.

Firefox quick-reference table

Setting	Location	Desired state
“Open previous windows and tabs”	about:preferences → Startup	Off
Enhanced Tracking Protection	about:preferences#privacy	Strict (or Custom)
Deceptive Content protection (4 boxes)	about:preferences#privacy	All On
HTTPS-Only Mode	about:preferences#privacy	Enable in all windows
Telemetry / studies / crash reports	about:preferences#privacy → Data Collection	All unchecked
Pocket / sponsored stories / sponsored shortcuts	about:preferences#home	All unchecked
Default search engine	about:preferences#search	Google / Bing / DuckDuckGo
DNS over HTTPS	about:preferences#privacy	Increased Protection
Built-in password manager	about:preferences#privacy → Logins and Passwords	Off
Background updater service	about:preferences#general → Firefox Updates	Off (on tight-RAM systems)
Profile count	about:profiles	Only profiles user knows about

Firefox URL shortcuts (appendix)

- [about:preferences](#) — main settings page
- [about:preferences#privacy](#) — tracking protection, HTTPS-Only, DoH, telemetry, password manager
- [about:preferences#home](#) — Pocket and sponsored content toggles
- [about:preferences#search](#) — search engine and search suggestions

- [about:preferences#general](#) — startup behavior and updates
- [about:addons](#) — review installed extensions, themes, plugins
- [about:profiles](#) — review installed Firefox profiles
- [about:performance](#) — per-tab and per-extension resource usage
- [about:memory](#) — detailed memory breakdown
- [about:support](#) — diagnostic info for troubleshooting

End of Firefox addendum.

Glossary

Definitions for key terms used throughout this guide. Useful for non-technical users following along.

2FA / Two-factor authentication — A login method that requires two pieces of evidence — typically a password plus a one-time code from an authenticator app, SMS, or email. Drastically reduces account takeover risk.

AppSuite — Publisher name behind a confirmed malware family (TamperedChef). Apps signed by AppSuite — especially generic “PDF Editor” products — should be treated as backdoors.

Authenticator app — A phone app that generates rotating one-time passcodes for 2FA. Examples: Microsoft Authenticator, Google Authenticator, Authy. More secure than SMS-based 2FA.

Backdoor — A type of malware that gives a remote attacker ongoing access to your machine, often hidden as a normal-looking program.

BitLocker — Microsoft’s built-in full-disk encryption for Windows. Protects data if the laptop is lost or stolen.

BitLocker recovery key — A 48-digit key Windows generates when BitLocker is enabled. Required to unlock the drive if Windows can’t. Must be stored OFF the encrypted drive.

Chromium — The open-source browser engine that powers Chrome, Edge, Brave, Opera, and many others. Apps that bundle Chromium (Slack, Discord, Teams) inherit its memory footprint.

Cloud-delivered protection — A Windows Defender feature that sends suspicious file fingerprints to Microsoft’s cloud for fast threat lookups. Should be On.

Controlled folder access — A Defender feature that blocks unauthorized apps from modifying files in protected folders (Documents, Pictures, etc.). Strong protection against ransomware.

Core Isolation / Memory Integrity — A Windows security feature that uses virtualization to isolate critical kernel processes from malware. Spawns vmmem and vmwp processes.

DDR4 / DDR5 — Generations of computer memory (RAM). DDR5 is newer and faster; older PCs use DDR4. Must match your motherboard.

Defender (Microsoft Defender) — Windows' built-in antivirus, firewall, and threat protection suite. Generally adequate to replace third-party AV on home PCs.

Defender Offline scan — A Defender mode that boots into a clean environment before Windows fully loads, allowing it to find rootkits and other malware that hide from regular scans.

DoH / DNS over HTTPS — Encrypts DNS lookups so your ISP and network can't see which websites you're asking for. Configurable in Firefox and Edge.

Edge — Microsoft Edge — Windows' default browser, based on Chromium.

Electron app — A category of desktop apps built on Chromium (e.g., Slack, Teams, Discord, Spotify, VS Code). Each typically uses 300–700 MB RAM.

Enhanced Tracking Protection (Firefox) — Firefox's feature blocking trackers, fingerprinters, and cryptominers. Three tiers: Standard, Strict, Custom.

Extended Security Updates (ESU) — Paid program letting Windows 10 users keep getting security patches after October 2025 end-of-support.

Firewall — Software that controls which network connections in and out of the PC are allowed. Windows includes one (Microsoft Defender Firewall).

Gigabyte (GB) — Unit of data/memory size. 1 GB = 1,024 MB. Modern Windows comfortably uses 8–16 GB of RAM.

Hyper-V — Microsoft's built-in virtualization technology. Memory Integrity, WSL2, Windows Sandbox, and WSA all depend on it.

Hypervisor / VBS — Virtualization-Based Security. Windows runs certain security functions inside a hypervisor-protected environment to isolate them from kernel-level malware.

MalwareBytes — Third-party anti-malware program known for catching PUPs and adware that Defender misses. Free version is sufficient.

Memory Integrity — See Core Isolation.

Memory Saver (Chrome) — Chrome feature that puts inactive tabs to sleep to free RAM. Settings: Moderate / Balanced / Maximum.

MPN / Model Part Number — Manufacturer's exact part identifier. Use this when buying replacement RAM (e.g., Crucial CT8G4SFRA32A) to ensure compatibility.

Page file — A hidden file Windows uses as overflow when RAM fills up. Heavy use indicates RAM is too small for the workload.

Pinned tab — A browser tab “locked” to the leftmost position. Pinned tabs reopen automatically every time the browser starts.

PowerShell — Windows' scripting and command shell. Often used for system administration tasks.

PUP / Potentially Unwanted Program — Software that isn't outright malware but is unwanted: scareware, fake cleaners, aggressive bundlers.

RAM / Memory — Random Access Memory — the fast, temporary storage your PC uses for running programs. More = better multitasking.

Ransomware — Malware that encrypts your files and demands payment for the key. Backups + Controlled Folder Access are key defenses.

Real-time protection — Defender's always-on scanning of files as they're opened, downloaded, or modified.

SO-DIMM — Small Outline Dual Inline Memory Module — the laptop form factor for RAM. Different from desktop DIMMs.

Startup boost (Edge) — An Edge feature that pre-launches Edge silently when Windows boots, so it opens faster. Should be Off if you don't want Edge auto-launching.

Tamper Protection — A Defender setting that prevents other software (including malware) from disabling Defender. Critical to keep On.

TamperedChef — Malware family distributed via fake utility apps (PDF Editor, ManualFinder, OneLaunch, Wave, Shift). Capable of password theft and full remote control.

Task Manager — Windows tool (Ctrl + Shift + Esc) for viewing running processes, CPU/RAM usage, and Startup apps.

Task Scheduler — Windows tool that runs programs on schedules. Edge and other apps add their own scheduled tasks here.

TPM / Trusted Platform Module — Hardware chip that securely stores encryption keys. BitLocker and Windows Hello PIN both rely on it.

UEFI / BIOS — Firmware that runs before Windows starts. Controls boot order, security features, and hardware initialization.

VBS — See Hypervisor.

VBR / VBS / vmem / vmwp — Processes related to Virtualization-Based Security. Memory Integrity uses ~150–300 MB across these processes; legitimate.

VMware / VirtualBox — Third-party virtualization software (run other operating systems inside Windows).

WebView2 — A Microsoft runtime that lets desktop apps embed web content using the Edge engine. Used by Outlook, Teams, Widgets, and many others.

Widgets (Windows) — The news/weather panel accessible from the taskbar. Uses Edge processes in the background, even if never opened.

Windows Defender — See Defender.

Windows Hello — Windows' biometric/PIN sign-in system. PIN is bound to the device's TPM and can't be stolen remotely.

Windows Update — Windows' automatic patching system. Critical for security; should always be current.

WSA / Windows Subsystem for Android — Microsoft's deprecated Android runtime for Windows 11. Spawns vmwp/vmmem if installed.

WSL / WSL2 — Windows Subsystem for Linux — runs a Linux distro inside Windows. Spawns vmwp/vmmem if a distro is running.

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